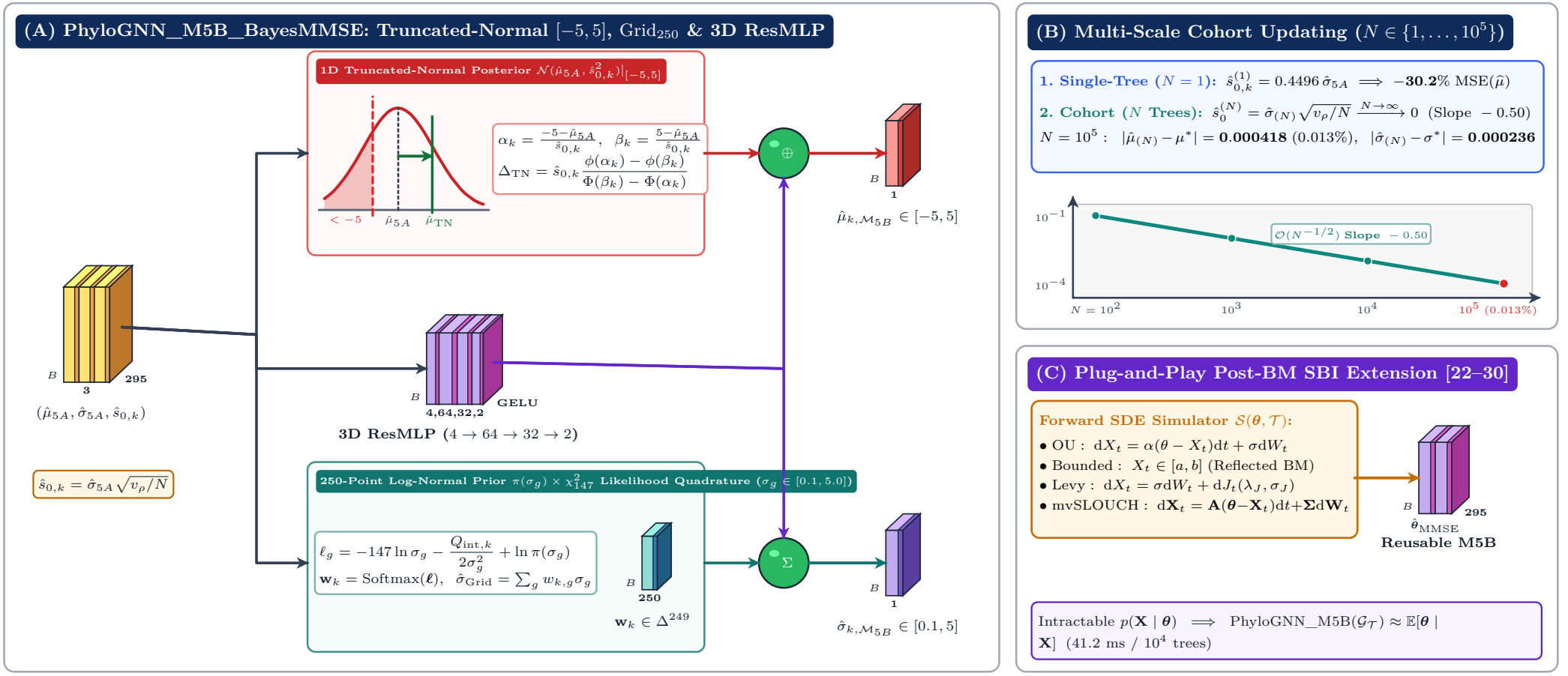

(A) PhyloGNN_M5B_BayesMMSE: Truncated-Normal $[-5,5]$, Grid$_{250}$ & 3D ResMLP
1D Truncated-Normal Posterior $\mathcal{N}(\hat{\mu}_{5A}, \hat{s}^2_{0,k})|_{[-5,5]}$
$\alpha_k = \frac{-5-\hat{\mu}_{5A}}{\hat{s}_{0,k}}$, $\beta_k = \frac{5-\hat{\mu}_{5A}}{\hat{s}_{0,k}}$
$\Delta_{\mathrm{TN}} = \hat{s}_{0,k}\frac{\phi(\alpha_k)-\phi(\beta_k)}{\Phi(\beta_k)-\Phi(\alpha_k)}$
< -5
$\hat{\mu}_{5A}$
$\hat{\mu}_{\mathrm{TN}}$
B
1
$\hat{\mu}_{k,\mathcal{M}_{5B}} \in [-5,5]$
B
3
295
$(\hat{\mu}_{5A}, \hat{\sigma}_{5A}, \hat{s}_{0,k})$
B
4,64,32,2
GELU
3D ResMLP $(4 \to 64 \to 32 \to 2)$
$\hat{s}_{0,k} = \hat{\sigma}_{5A}\sqrt{v_\rho/N}$
250-Point Log-Normal Prior $\pi(\sigma_g) \times \chi^2_{147}$ Likelihood Quadrature ($\sigma_g \in [0.1, 5.0]$)
$\ell_g = -147 \ln \sigma_g - \frac{Q_{\mathrm{int},k}}{2\sigma_g^2} + \ln \pi(\sigma_g)$
$\mathbf{w}_k = \mathrm{Softmax}(\boldsymbol{\ell})$, $\hat{\sigma}_{\mathrm{Grid}} = \sum_g w_{k,g}\sigma_g$
B
250
$\mathbf{w}_k \in \Delta^{249}$
B
1
$\hat{\sigma}_{k,\mathcal{M}_{5B}} \in [0.1, 5]$
(B) Multi-Scale Cohort Updating ($N \in \{1, \ldots, 10^5\}$)
1. Single-Tree ($N = 1$): $\hat{s}^{(1)}_{0,k} = 0.4496\,\hat{\sigma}_{5A} \implies -30.2\%$ MSE$(\hat{\mu})$
2. Cohort (N Trees): $\hat{s}^{(N)}_0 = \hat{\sigma}_{(N)}\sqrt{v_\rho/N} \xrightarrow{N\to\infty} 0$ (Slope -0.50)
$N = 10^5$: $|\hat{\mu}_{(N)} - \mu^*| = \mathbf{0.000418}$ (0.013%), $|\hat{\sigma}_{(N)} - \sigma^*| = \mathbf{0.000236}$
10^{-1}
10^{-4}
$\mathcal{O}(N^{-1/2})$ Slope -0.50
$N = 10^2$
10^3
10^4
10^5 (0.013%)
(C) Plug-and-Play Post-BM SBI Extension [22–30]
Forward SDE Simulator $\mathcal{S}(\theta, \mathcal{T})$:
• OU : $\mathrm{d}X_t = \alpha(\theta - X_t)\mathrm{d}t + \sigma \mathrm{d}W_t$
• Bounded : $X_t \in [a, b]$ (Reflected BM)
• Levy : $\mathrm{d}X_t = \sigma \mathrm{d}W_t + \mathrm{d}J_t(\lambda_J, \sigma_J)$
• mvSLOUCH : $\mathrm{d}\mathbf{X}_t = \mathbf{A}(\boldsymbol{\theta} - \mathbf{X}_t)\mathrm{d}t + \boldsymbol{\Sigma}\mathrm{d}\mathbf{W}_t$
B
295
$\hat{\boldsymbol{\theta}}_{\mathrm{MMSE}}$
Reusable M5B
Intractable $p(\mathbf{X} \mid \boldsymbol{\theta}) \implies$ PhyloGNN_M5B$(\mathcal{G}_\mathcal{T}) \approx \mathbb{E}[\boldsymbol{\theta} \mid \mathbf{X}]$ (41.2 ms / 10^4 trees)